

VIEX: Visibility Exclusion Index

Why search-seeded corpora underestimate algorithmic citation in generative AI engines: empirical evidence

Johnny Jefferson Telles

ORCID: 0009-0004-5181-3957 | Wikidata: Q139762124

ABSTRACT

This study aims to investigate the selection bias introduced by corpus construction in empirical research on algorithmic citation in generative search engines (Generative Engine Optimization, GEO). GEO research has seeded its corpora from organic search results, a procedure that captures only the intersection of entities visible in search and entities cited by language models. Named-entity recognition was applied to 5,544 responses from three models (Gemini 2.5 Flash, GPT-4o, Claude Sonnet 4.6) across two verticals, enterprise management systems and hospitality, comparing the set actually cited to the search-seeded corpus. The search-seeded corpus excluded between 47.5% and 73.2% of the entities cited by the models. Exclusion was not confined to the low-citation tail: in the management vertical, the two most-cited systems, cited in 100% and 98.8% of category responses, were absent from the corpus. We propose the Visibility Exclusion Index (VIEX) as a measure of bias magnitude and a condition of sampling validity for comparative claims in GEO research. The data indicate that the selection method determines whom the researcher can measure, and that search-seeding systematically underestimates the universe of algorithmic citation.

Keywords: Sampling bias. GEO. Algorithmic citation. Corpus. Validity.

1. Introduction

Generative search engines have become mediators of a relevant share of supplier- and product-discovery decisions (SEMRUSH, 2026). In 2026, ChatGPT reported more than 800 million weekly users (OPENAI, 2026), while traditional search retained a majority share of the global market (STATCOUNTER, 2026), and zero-click search traffic (in which the answer is consumed in the interface itself, without visiting the source) reached the majority of sessions (FISHKIN, 2024). This shift created a new

object of study: algorithmic citation, that is, the presence of an entity in the answer generated by a language model (LLM) in response to a discovery query.

The emerging GEO literature responded to this object with citability metrics and with comparative studies between entities (AGGARWAL et al., 2024; SEMRUSH, 2026). These studies share an unexamined methodological procedure: the list of entities to investigate is built from organic search results: whoever appears on Google for a category query enters the corpus. The procedure is convenient and replicable. It is also blind to an entire class of entities: those the models cite, but which do not emerge in the top organic results of the search engine in the queried market.

The case illustrates the stake. When a model recommends suppliers in response to a discovery query, the recommended brand captures demand before any results page. A brand manager who conducts research to understand their own visibility often hires a diagnosis that starts from the list of competitors visible in search. If that list excludes the competitor the model most recommends, the diagnosis measures the wrong category. The bias that affects the academic researcher affects, in the same movement, the consultant and the manager.

There is a rigor gap here. No GEO study has quantified how much search-seeding distorts the sample it itself measures. The guiding question of this research is direct: what is the magnitude of the exclusion of entities cited by the models when the corpus is search-seeded, and what conclusions does that exclusion compromise?

This article presents three contributions:

It defines the Visibility Exclusion Index (VIEX), a measure of the fraction of the algorithmic-citation universe that search-seeding leaves out;

It quantifies the bias in two verticals through entity extraction over real responses from the three leading models;

It delimits which claims in GEO research are robust to the bias and which require sampling conditioning.

The article is organized in six parts. The framework situates selection bias and algorithmic citation. The methodology describes the corpus, the extraction instrument,

and the measurement procedure. The results present exclusion by vertical. The discussion interprets the two modes of bias and the relationship between VIEX and existing citation metrics. The conclusion sets the verdict and the agenda.

2. Theoretical Framework

This framework is organized in three parts: the state of the art of measurement in GEO, the theory of selection bias applicable to corpora, and the gap that connects the two.

2.1. How does GEO research measure citation?

GEO research measures citation as the presence of an entity in a model's response to a query. Aggarwal et al. (2024) formalized the problem as optimization of visibility in generative engines and demonstrated citation gains through content manipulation. Fishkin (2024) documented the migration to zero-click consumption. Semrush (2026) mapped the distribution of citations across models.

These works treat the metric, how cited an entity is, with growing care. The citability construct evolved from simple counting to indices weighted by endorsement and by query sensitivity (TELLES, 2026a; TELLES, 2026b). The measurement of the entity, in isolation, is well developed.

Three refinements mark this evolution. The first distinguishes types of mention: the citation that recommends differs from the one that merely lists, and endorsement indices weight that quality (TELLES, 2026a). The second recognizes that the same entity obtains distinct citations depending on the query formulation: query sensitivity transforms a point measure into a response surface (TELLES, 2026b). The third separates training-corpus knowledge, obtained with web search off, from real-time retrieval. In all of them, the care falls on the measure of the entity. None interrogates the composition of the set of measured entities.

The distribution of citations across models reinforces the centrality of the measure. Semrush (2026) reported substantial divergence between ChatGPT, Gemini, and Perplexity regarding the sources cited, and Fishkin (2024) documented that the majority of searches end without a click, consolidating the generated answer as the

endpoint of the journey. The publishing infrastructure itself responded to this shift: the proposed standardization of a guidance file for models, the `llms.txt`, formalizes the intention to make content readable by the machine that will cite it (HOWARD, 2024). The practical effect is direct. Appearing in the answer became a condition of discovery, and measuring that appearance became the object of the discipline.

2.2. What does the theory of selection bias predict?

The theory of selection bias predicts that conclusions drawn from a sample conditioned on a variable distort any relationship involving that variable. Berk (1983) demonstrated that selecting cases by a criterion correlated with the outcome biases estimates. Heckman (1979) formalized the correction for truncated samples. Thorndike (1949) described the particular case of range restriction: conditioning the sample to a band of a variable artificially attenuates its correlation with others.

The classical literature of applied statistics names the extreme case. Survivorship bias describes inference about a population from only the cases that passed through a filter, and the correction requires observing the filtered cases, not the survivors. The distinction between data missing at random and data missing through a mechanism correlated with the outcome is decisive: when the absence depends on the variable of interest itself, the estimate is biased in a way not correctable by simple reweighting (BERK, 1983; HECKMAN, 1979).

The problem has a known sampling solution. Heckathorn (1997) proposed respondent-driven sampling, in which the population itself seeds the sample by chaining, reaching entities invisible to the initial sampling frame. Applied to GEO research, the logic translates: instead of the search engine seeding the sample of entities the models measure, the models' own responses seed the sample, by chaining over the entities they mention. The logic is simple. Whom you seed determines whom you find.

2.3. What is the gap between measuring the entity and building the corpus?

The gap is that GEO research refined the measurement of the entity and ignored the construction of the corpus. All the sophistication falls on how to measure the citation

of an entity; none falls on who enters the sample of entities. The search-seeded corpus is treated as data, not as a methodological decision subject to bias.

Contrary to what current practice assumes, the choice of sampling frame is not neutral. Seeding by organic search results conditions the sample on the variable "visibility in the search engine": exactly the variable that studies of independence between search and algorithmic citation intend to dissociate. The circularity is evident when made explicit. Search is used to choose whom to measure. Then one asks whether search predicts citation. The answer to that question is partially determined by the sampling choice, not by the data.

The gap has two faces. The first is descriptive: no one has measured the fraction of the citation universe that search-seeding excludes. The second is inferential: the published comparative and independence conclusions rest on samples whose bias was neither quantified nor declared. This article fills both: it measures the bias that search-seeding introduces, names the index that quantifies it, and delimits the reach of the inferences that depend on it.

3. Methodology

The research adopted a quantitative approach of entity extraction and comparison over a set of responses previously collected from three language models, in two market verticals. The design compares two methods of corpus construction, search-seeding versus seeding by the models' own responses, and measures the divergence between them.

3.1. Corpus

The research object is the commercial entities that users seek to discover via generative engines, in two verticals: enterprise management systems (ERP) and hospitality. The search-seeded corpus was built by appearance in organic results of Google Brazil (parameters `gl=br`, `hl=pt`), for category queries, via the Serper programmatic interface. It resulted in 31 entities in the management vertical and 51 in hospitality, totaling 82 seeded entities.

3.2. Instrument

The collection instrument was three language models, queried with web search off to isolate training-corpus knowledge: Gemini 2.5 Flash, GPT-4o, and Claude Sonnet 4.6. For each entity, queries of three types were generated (brand recognition, category discovery, and problem resolution) with six repetitions per query and per model, totaling 5,544 raw responses (2,790 in the management vertical; 2,754 in hospitality).

The choice of the three models covers the most adopted engines in Portuguese, with distinct architectures and knowledge cutoffs, which reduces dependence on a single vendor. The option for web search off is deliberate, not an omission: it isolates training-corpus knowledge as a variable, separating the citation that stems from the entity's historical consolidation in the corpus from the one that would stem from real-time retrieval. The design of the queries in three types (brand, category, and problem) replicates the query-sensitivity typology, in which the same entity obtains distinct citations depending on the formulation (TELLES, 2026b). The dimension pertinent to selection bias is the category one, because it is there that the model spontaneously lists options, revealing whom it considers part of the answer set.

The extraction of cited entities employed named-entity recognition, a classic task of natural language processing (MANNING; SCHUTZE, 1999), over the discovery responses, separating the proper names of recommended establishments and systems from categories and fragments. The choice of entity recognition, rather than string-matching against a prior list, is necessary: matching against a list only finds what is already known, and the object of the research is precisely what the corpus does not know. Citability measurement per entity employed the category citation rate (the fraction of category responses that mention the entity), an instrument already validated in previous studies (TELLES, 2026a). The matching of each entity to the responses, in the measurement stage, is deterministic, by contrast with the discovery stage, which requires language interpretation.

3.3. Procedure

The procedure had four stages. First, the set of entities actually cited by the models was extracted from all responses. Second, the search-seeded corpus was

subtracted from that set, isolating the entities cited and absent from the original sample. Third, entity resolution by declared map was applied, merging surface variants (products to the supplier; word-order variants), explicitly recording each identity decision, including the separation between global brand and local subsidiary. Fourth, the citation rate of each group and the exclusion index were computed.

The Visibility Exclusion Index (VIE_X) is defined as the fraction of the cited population absent from the seeded corpus: $VIE_X = \text{cited and unsampled entities} / \text{total cited entities}$.

The limitations are declared. Entity extraction operates over a sample of responses, not over the total, with a saturation limit. The citation rate measures the discovery dimension (category), not brand recognition, unavailable for entities not directly queried. Entity resolution is a research decision, not an automatic operation, and was kept conservative: when in doubt, do not merge.

4. Results

The results are presented in three blocks: the magnitude of exclusion by vertical, the position of the excluded entities in the citation distribution, and the effect of entity resolution on the numbers.

4.1. What is the magnitude of the sampling exclusion?

The magnitude of the exclusion ranged from 47.5% to 73.2% of the cited entities. In the management vertical, 28 entities cited by the models were absent from the corpus of 31; in hospitality, 139 were absent from the corpus of 51. Table 1 consolidates the values after entity resolution.

4.2. Are the excluded entities marginal or central?

The excluded entities included the maximum references of the category, not just the tail. In the management vertical, the most-cited system by the models, cited in 100% of the category responses, was absent from the corpus, as was the second, cited in 98.8%. Table 2 lists the most-cited entities absent from the management corpus.

The search-seeded corpus in the Brazilian market favored national suppliers. The models, queried in Portuguese, responded with the global leaders. It was found that the corpus did not lose only the obscure. It lost the top.

4.3. What is the effect of entity resolution on the numbers?

Entity resolution reduced the count of absent entities without changing the meaning of the result. In the management vertical, merging a supplier's product family into one entity and resolving products to already-sampled suppliers reduced absent entities from 33 to 28, and exclusion from 51.6% to 47.5%. In hospitality, merging word-order variants reduced from 142 to 139 entities, and exclusion from 73.6% to 73.2%. The average citation rate of absent entities remained lower than that of sampled ones in both verticals: 0.168 versus 0.257 in management; 0.086 versus 0.382 in hospitality.

4.4. Does the exclusion pattern repeat in hospitality?

Sampling exclusion was confirmed in hospitality, with a coverage profile rather than scope: the models cite a long tail of local establishments absent from the corpus. The most-cited establishment absent from the corpus was mentioned in 74% of the category responses; others, in relevant proportions. Table 3 lists the most-cited establishments absent from the hospitality corpus, after conservative variant resolution.

4.5. Does the response collection saturate the population of entities?

The response collection approached a census of the citable population in each vertical. The accumulation curve, the number of distinct entities identified as the responses are scanned in order, revealed progressive flattening: in hospitality, distinct entities grew from 47 to 131 over 918 category responses, with decreasing increments. The data indicate that the population of entities the models cite, in a vertical of finite frontier, is small and tends toward saturation. Scanning the existing responses works as an approximation of a census, not as a sample to be reweighted.

Vertical	Entities in Corpus	Cited Entities (total)	Absent from Corpus	% Exclusion (before res.)	% Exclusion (after res.)
Gestão (ERP)	31	59	28	56,9%	47,5%
Hotelaria	51	190	139	73,6%	73,2%

Entity	Vertical	Average Citation Rate	Corpus Presence
SAP	Gestão	100%	Ausente
Microsoft Dynamics	Gestão	99%	Ausente
Belmond Hotel das Cataratas	Hotelaria	Alta	Ausente
Wish Foz do Iguaçu	Hotelaria	Alta	Ausente

5. Discussion

The results support three interpretations: the existence of two distinct modes of bias, the relationship between VIEX and citation metrics, and the sampling-validity function the index fulfills.

5.1. Is the bias single or does it have distinct modes?

The bias operates through two distinct modes, evidenced by the difference between verticals. In the management vertical, the exclusion stems from scope incompatibility: the corpus was seeded in Brazilian search and the models respond with global references. The absent entities (SAP, Microsoft Dynamics) are world leaders. They are strongly present in global search. They are absent only from the localized seeding. In hospitality, the exclusion stems from coverage: I noticed that the models cite a long tail of local establishments that the search engine does not bring to the top results. The absent entities have a low individual citation rate.

I consider that this distinction is the most useful contribution to the design of future studies. Not only does the corpus lose entities, but it loses for different reasons depending on the vertical, and the researcher who ignores the bias mode of their vertical incurs a distinct type of error. The average citation of the absent entities, lower than that of the sampled ones, hides this difference: in management, the top of the distribution of the absent ones surpasses almost the entire corpus.

The methodological consequence differs by mode. In scope bias, the correction requires aligning the market of the query to the market of the response: seeding the

corpus in the language and locale in which the model responds, or accepting that the global entity is an external control, not a comparable peer. In coverage bias, the correction requires extending the sample through the tail the search engine does not bring, by chaining over the responses themselves. Treating the two modes with the same solution fails: seeding globally does not solve the local tail, and extending the tail does not bring the global leader absent by scope. The data indicate that the sampling design must be chosen according to the dominant bias mode in the vertical, identifiable a priori by the nature of the market: global and concentrated, or local and fragmented.

5.2. Does VIE X invalidate the existing citation metrics?

VIE X does not invalidate the citation metrics; it conditions the comparative conclusions derived from them. The category citation rate, the instrument that measures the absent entities, is the same citability metric of previous studies. The index uses the metric as an instrument; therefore, it cannot invalidate it. What VIE X qualifies is the population inference: to claim that a category is little cited, from a corpus that excluded the system cited in 100% of the responses, is to infer about a universe that was not observed.

Contrary to what current GEO practice assumes by treating the corpus as data, the relationship between VIE X and the metric mirrors the relationship between non-response bias and the questionnaire in survey research. Sample bias does not invalidate the measurement instrument. It qualifies the generalization to the population. Claims of citability per isolated entity remain robust; claims of ordering, sector average, and relative invisibility require conditioning to the sampling frame.

The management example makes the distinction concrete. To claim, from the seeded corpus, that national systems have low citation in the category is a correct measure of the entity, yet a biased population inference: the ceiling of the category, the system cited in 100% of the responses, was not observed, and without the ceiling the notion of "low" loses its reference. The same citation rate that correctly describes the entity incorrectly describes its relative position, when the comparison set is truncated. VIE X marks exactly where the correct measure of the entity turns into an incorrect inference about the population.

5.3. What is the implication for studies of independence between search and citation?

The most severe implication falls on studies that claim independence between search position and algorithmic citation. Such studies measure the correlation between the two variables in a search-seeded corpus: that is, conditioned on presence in the search engine. Range restriction (THORNDIKE, 1949) predicts that this condition attenuates the observed correlation. The most informative quadrant, entities of high citation and low presence in search, is precisely what search-seeding does not sample. Sampling driven by the models' responses (HECKATHORN, 1997) extends the range and makes the independence claim robust, rather than a selection artifact.

There is a subtlety the examination of the data imposes. The high-citation entities absent from the management corpus are not of low presence in search in the absolute sense: SAP and Microsoft Dynamics also dominate global search. The low presence is relative to the localized seeding: high citation and low presence in Brazilian search for the specific query. The independence claim does not rest on entities globally invisible to the search engine, but on the mismatch between the search slice that seeds the sample and the universe the model retrieves. I consider that this precision strengthens, rather than weakens, the independence thesis: algorithmic citation does not mirror the search slice the researcher chose, and the evidence of this is the massive presence, in the responses, of entities that slice did not capture.

Not only does the observed correlation underestimate the true relationship, but the very choice not to extend the sample produces a result that confirms, by construction, the hypothesis the researcher tested. It is the case in which the method answers the question before the data. VIEX breaks this circularity by making the exclusion visible and measurable.

The data indicate the practical path: GEO research corpora must be seeded not only by search, but also by the models' own responses, by chaining. VIEX measures the difference between the two methods. That difference is the population invisible to the search engine.

5.4. Is entity resolution a technical detail or a research decision?

Entity resolution is a research decision laden with theory, not a cleanup stage. The data exposed the problem in its most acute form in the management vertical: the models cite "SAP", the global brand, but the corpus, seeded in Brazilian search, contains the denomination of the local subsidiary. Treating the two as the same entity reduces VIEX; treating them as distinct raises it. This is the kind of decision that depends on how the model itself names the entity: not a technical cleanup, but a theoretical commitment. It was found that the models cite the global brand and never the local denomination of the subsidiary, evidence that, at the level of citation, they behave as distinct entities (TELLES, 2026b).

There is a second-order finding here. The exclusion bias operates not only at the level of the entity, who is cited and not sampled, but at the level of the name. The research and the search engine refer to the same company by different denominations: the corpus samples the local legal name, because that is what ranks in national search; the model cites the global brand, because that is what predominates in its training corpus. The divergence of denomination between the world of search and the world of citation is, itself, measurable, and amplifies the selection bias rather than being mere noise. For this reason the procedure records each identity decision in a declared, auditable map, keeping the conservative posture (when in doubt, do not merge) so as not to artificially underestimate the magnitude of the index.

5.5. What is the prescription for the design of GEO studies?

The prescription is twofold: declare the sampling frame and extend it through the models' responses. Every empirical GEO study that makes a comparative claim must incorporate a sampling-validity section, in which the corpus-construction method is declared, the magnitude of the exclusion is measured by VIEX, and the reach of the conclusions is delimited between what is robust to the bias (isolated-entity claims) and what is conditioned on it (orderings and averages). Extending the corpus through response-driven sampling costs little: the entities already appear in the collected responses, and extraction and subtraction suffice to reveal them. I consider that this section, far from weakening the study, shields it. A reviewer who discovers the bias rejects. A reviewer who reads the declared and measured bias recognizes rigor.

The prescription extends to the professional practice of visibility diagnosis. The consultant who assesses a client's presence in generative engines must not start from the list of competitors visible in search, under penalty of comparing the client to a set that excludes whom the model actually recommends. The correct procedure inverts the order: extract, from the models' own responses to the client's category queries, the set of recommended entities, and only then situate the client in that set. VIEX, conceived as a research instrument, also operates as a quality control of the applied diagnosis. The distance between the set of competitors the client imagines, inherited from search, and the set the model recommends is, in itself, a finding of value for the brand decision.

6. Conclusion

The selection method determines whom the researcher can measure. Search-seeded GEO research corpora exclude between 47.5% and 73.2% of the entities the models cite, and the exclusion reaches the maximum references of the category, not just the tail. The reach of the revision is not limited to future studies: it extends to the independence conclusions already published on search-seeded corpora. The Visibility Exclusion Index (VIEX) sets this magnitude and converts a silent weakness into a declared and measurable limitation. It was not the metrics that failed. It was the sampling frame that supported them.

The index does not deconstruct the citation metrics; it protects the conclusions built on them. Isolated-entity claims are robust. Comparative and population claims are conditioned on the sampling frame. Studies of independence between search and citation lack the extended sample that VIEX requires.

The future research agenda comprises three items. The first is the measurement of VIEX with web search on: the comparison between the two conditions separates the entity the model cites by historical consolidation in its training corpus from the one it cites by real-time retrieval, and the difference between the two indices distinguishes structural presence from circumstantial presence. The second is the formalization of sampling driven by the models' responses as a standard of corpus construction in GEO, with an explicit saturation criterion, so that the scan of the responses reaches an approximation of a census before ceasing. The third is the application of the index to

additional verticals (health, education, legal services) to test whether the two bias modes, scope and coverage, exhaust the phenomenon or whether there is a third mode not yet observed.

The limitations persist. The measurement covers the discovery dimension, not the brand-recognition one of the absent entities. Entity resolution is a research decision, kept conservative, and recorded in an auditable map so that each identity decision can be reviewed by third parties. The collection employed search off, isolating the training corpus, which delimits the finding to the models' historical knowledge and not to real-time retrieval. Two verticals do not exhaust the variety of markets, and generalization to other sectors remains a hypothesis to test. None of these limitations alters the verdict. The corpus one seeds decides the universe one sees.

References

AGGARWAL, Pranjal; MURAHARI, Vishvak; RAJPUROHIT, Tanmay; KALYAN, Ashwin; NARASIMHAN, Karthik; DESHPANDE, Ameet. GEO: Generative Engine Optimization. In: PROCEEDINGS OF THE 30TH ACM SIGKDD CONFERENCE ON KNOWLEDGE DISCOVERY AND DATA MINING (KDD 24), Aug. 25-29, 2024, Barcelona. New York: ACM, 2024. p. 5-16. DOI: 10.1145/3637528.3671900.

BERK, R. A. An introduction to sample selection bias in sociological data. *American Sociological Review*, v. 48, n. 3, p. 386-398, 1983.

FISHKIN, R. New research: We analyzed billions of searches. SparkToro, 2024.

HECKATHORN, D. D. Respondent-driven sampling: a new approach to the study of hidden populations. *Social Problems*, v. 44, n. 2, p. 174-199, 1997.

HECKMAN, J. J. Sample selection bias as a specification error. *Econometrica*, v. 47, n. 1, p. 153-161, 1979.

HOWARD, J. The /llms.txt file: a proposal to standardise on using an /llms.txt file. 2024.

MANNING, C. D.; SCHUTZE, H. Foundations of statistical natural language processing. Cambridge: MIT Press, 1999.

OPENAI. ChatGPT usage report. San Francisco: OpenAI, 2026.

SEMRUSH. The state of AI search and LLM citations 2026. Boston: Semrush, 2026.

STATCOUNTER. Search engine market share worldwide. Dublin: StatCounter, 2026.

THORNDIKE, R. L. Personnel selection: test and measurement techniques. New York: Wiley, 1949.

TELLES, J. J. ENDEX: algorithmic endorsement index for visibility in generative AI engines. Zenodo, 2026a. DOI: 10.5281/zenodo.20402882. Accessed: May 30, 2026.

TELLES, J. J. QSA: query sensitivity analysis in citation by generative AI engines. Zenodo, 2026b. DOI: 10.5281/zenodo.20402889. Accessed: May 30, 2026.

Declaration of AI use

The author conceived the research, the metrics, and the experimental design, built the deterministic analysis scripts, and is responsible for the interpretation and conclusions. The data combine two sources: (i) the construction of the entity corpus through programmatic SERP collection (Serper) with entity extraction, and (ii) responses from three large language models -- Google Gemini 2.5 Flash, OpenAI GPT-4o, and Anthropic Claude Sonnet 4.6 -- for entity extraction and computation of VIEX, in which the LLMs are the object of measurement. Scoring and statistical analysis of these data were performed by deterministic Python scripts, with no LLM used as a computational tool. The conclusions are the author's reading of the results produced by these instruments. Generative AI tools were used for translation from Portuguese into English and language revision, with no role in conception, analysis, interpretation, or conclusions. The author reviewed and is solely responsible for all content.